Output Form (OF)

Score: 3/5 — Moderate gaps | Confidence: medium

Benchmark: *WXIMPACTBENCH* | ***Context:*** *Argentine Government Disaster Reporting Reliability — Policy Staff*

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Output form consists of binary label vectors and ranked article lists [Q33, Q36, Q43–Q45, Q50], producing F1, accuracy, row-wise accuracy, Hit@1, nDCG@5, Recall@5, and MRR. These outputs are usable as upstream features for the deployment's downstream credibility aggregation per A3, so the form is functionally compatible. However, the benchmark does not natively produce the credibility/reliability score that constitutes the deployment's primary decision output, and the transformation from binary labels to credibility scores requires non-trivial integration with Argentine source metadata (outlet reach, account verification, institutional affiliation) that is unaddressed. Row-wise accuracy [Q45, Q75] provides a stricter evaluation that more closely approximates aggregation reliability needs. Output form is MODERATE-priority per elicitation; score reflects functional compatibility tempered by the non-trivial transformation gap and the absence of confidence/probability outputs that would be most useful for downstream aggregation.

ID	Evidence
Q33	"Specifically, given an article sample xt corresponding to six ground-truth impacts Yt = {y^i_t}^6_i=1 with binary labels y^i_t ∈ {0, 1}, the evaluated model M is required to maximize the probability of the predicated impact Ŷ_t = {ŷ^i_t}^6_i=1 towards ground-truth." (p.5)
Q36	"We formulate the ranking-based QA task by prompting the models to identify the likelihood of each article containing the correct answer from a candidate pool." (p.5)
Q43	"For multi-label classification task, we use F1-score, accuracy, and row-wise accuracy as evaluation metrics." (p.6)
Q44	"The evaluation via F1-score and accuracy are averaged across the six impact categories, historical and modern articles, and the effect of different context lengths." (p.6)
Q45	"Compared to the common F1-score and accuracy, the row-wise accuracy is a strict metric that requires more accurate output as the model should correctly classify all six impact labels for a given article." (p.6)
Q50	"For the ranking-based QA task, we deploy the standard metric that emphasizes the accuracy of top positions for evaluation, including Hit@1, nDCG@5, Recall@5, and MRR." (p.6)
Q75	"Row-wise accuracy requires an exact match across all labels." (p.8)

Strengths

- Binary label vectors and ranked lists are functionally compatible as upstream features for credibility aggregation per A3
- Multiple metrics including stricter row-wise accuracy [Q45] capture different aspects of reliability
- Standard IR metrics (Hit@1, nDCG@5, Recall@5, MRR) [Q50] for the QA task align with retrieval-pipeline evaluation
- Modest computational requirements [Q127, Q128] enable replication on Argentine-relevant content

ID	Evidence
Q45	"Compared to the common F1-score and accuracy, the row-wise accuracy is a strict metric that requires more accurate output as the model should correctly classify all six impact labels for a given article." (p.6)
Q50	"For the ranking-based QA task, we deploy the standard metric that emphasizes the accuracy of top positions for evaluation, including Hit@1, nDCG@5, Recall@5, and MRR." (p.6)
Q127	"For all open-source models, evaluations were performed on a system with two NVIDIA A6000 (32GB) GPUs." (p.17)
Q128	"The relatively modest computational requirements highlight the accessibility of our benchmark for researchers with limited computational resources, while still enabling comprehensive evaluation of state-of-the-art models" (p.17)

Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Output modality (binary labels + rankings) is text-based and compatible with deployment's text channels; functional for upstream feature use [A3]. However, native credibility/reliability score is absent — the user accepts this and plans downstream aggregation [A3].

OF-2: Text-to-speech is not relevant: target users are policy professionals consuming text-based reports [target_population description]; literacy is near-universal (~99%) [WEB-2, WEB-3] and not a constraint.

OF-3: Literacy and education are not limiting factors (target population holds advanced degrees; national literacy ~99% [WEB-2, WEB-3, WEB-4]). Output form's text-only nature is appropriate.

OF-4: External validity gap: benchmark does not produce probability scores or confidence outputs that would be most useful for downstream credibility aggregation; only binary labels and discrete rankings. Integration with Argentine source metadata (outlet, account verification, institutional affiliation) is not addressed within benchmark scope.

ID	Evidence
Q33	"Specifically, given an article sample xt corresponding to six ground-truth impacts Yt = {y^i_t}^6_i=1 with binary labels y^i_t ∈ {0, 1}, the evaluated model M is required to maximize the probability of the predicated impact Ŷ_t = {ŷ^i_t}^6_i=1 towards ground-truth." (p.5)
Q43	"For multi-label classification task, we use F1-score, accuracy, and row-wise accuracy as evaluation metrics." (p.6)
WEB-2	Argentina national adult literacy ~99% (https://countryeconomy.com/demography/literacy-rate/argentina)
WEB-3	World Bank/UNESCO confirms literacy >98% since 2001 (https://www.macrotrends.net/global-metrics/countries/arg/argentina/literacy-rate)
WEB-4	OECD 2025: 19% tertiary qualification among 25–34 year-olds; target cohort holds advanced degrees (https://gpseducation.oecd.org/CountryProfile?primaryCountry=ARG&treshold=10&topic=EO)

Information Gaps

- Whether models can produce calibrated probability outputs in addition to binary labels for downstream credibility aggregation
- Specification of the Argentine source-metadata schema for credibility aggregation

Requires Expert Verification

- Design of the credibility aggregation layer that combines benchmark outputs with Argentine source metadata

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Remediation

Gap 1: Binary labels only; no calibrated probability outputs for downstream credibility aggregation [Q33]

Recommendation: Re-run benchmark evaluation extracting model log-probabilities or calibrated confidence scores per label rather than binary outputs. Evaluate whether confidence calibration on WXImpactBench correlates with calibration on the Argentine test set built under remediation 1, to inform the credibility aggregation layer.

ID	Evidence
Q33	“Specifically, given an article sample xt corresponding to six ground-truth impacts Yt = {y^i_t}^6_i=1 with binary labels y^i_t ∈ {0, 1}, the evaluated model M is required to maximize the probability of the predicated impact Ŷ_t = {ŷ^i_t}^6_i=1 towards ground-truth.” (p.5)